Black Soldier fly (*Hermetia illucens*) larvae in waste valorization and diet-based shifts in their gut microbiome

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Aims

Highlighting the Black Soldier Fly as an efficient all-round degrader at the:

Larval level
Optimal ratios of different organic wastes
↓
Maximize degradation and metabolization by larvae
↓
Highest possible increase in larval biomass and a minimum of substrate residues

Microbial level
Investigate microbial gut communities
↓
Assess influence of different diets on predominant microbiota
↓
What role do they play in larval activity?

Introduction

Fast-moving **consumerism** and **deterioration** of food products

Increase of municipal biodegradable waste in the EU:
14 Mt in **1995** to **40 Mt** in **2016**
(*Eurostat*, 2018)

CLASSIC WAYS
Deposition in landfills
Aerobic **degradation** (compost)
Anaerobic **fermentation** (biogas)

ALTERNATIVE WAY
Conversion of organic waste to **INSECT BIOMASS**
(*Wang and Shelomi*, 2017)

Industrial building blocks:
Proteins
Lipids
Chitin

Animal feed:
Fish
Pigs
Poultry

Methods

Rearing conditions for larvae:

- Environmental conditions set to 27 °C and 60% relative humidity
- Kept in separate non-transparent plastic boxes (25 x 10 x 5 cm; four replicates per treatment)
- 200 six-day old larvae per box as starting point for the experiment
- Feed amounts: 10, 100, and 130 mg larva^{-1} day^{-1} for chickenfeed, fruit, and grass, respectively

Sampling and sample preparation:

- 20 larvae were collected every three days from each box, and dry weight, organic and anorganic content were determined
- Guts were removed from surface sterilized (70% EtOH) larvae, and DNA was extracted using the NucleoSpin® Soil-Kit (Macherey-Nagel GmbH, GER) after homogenization

Analysis of Illumina Amplicon sequencing data:

- Sequencing was performed by Microsynth AG (Balgach, CH), and raw reads were analyzed in mothur (*Schloss et al.*, 2009)
- Non metric multidimensional scaling (NMDS) was calculated using vegan (*Oksanen et al.*, 2016) in R (*R Development Core Team*, 2008)

Results

Tab. 1: Mean larval mass balance from diets based on organic wastes and chickenfeed for comparison. The WRI describes the relation between overall degradation and time (higher is better). The EDC explains how efficiently larvae transform the offered substrate into biomass (higher is better).

	Organic wastes (fruit, vegetables, grass cuttings)		Chickenfeed	
	Mean	SE	Mean	SE
Residuals	31.0%	10.0%	27.8%	6.9%
Biomass	7.6%	2.4%	20.2%	2.3%
Metabolism	61.5%	1.3%	51.9%	0.6%
Waste Reduction Index (WRI)	1.4	0.1	1.7	0.1
Efficiency of Conversion of Digested Feed (ECD)	21.0%	1.8%	49.8%	2.0%
Actinobacteria	25.6%	11.9%	48.8%	18.9%
Bacteroidia	31.6%	18.4%	25.3%	20.3%
Proteobacteria	27.2%	19.4%	14.1%	11.4%

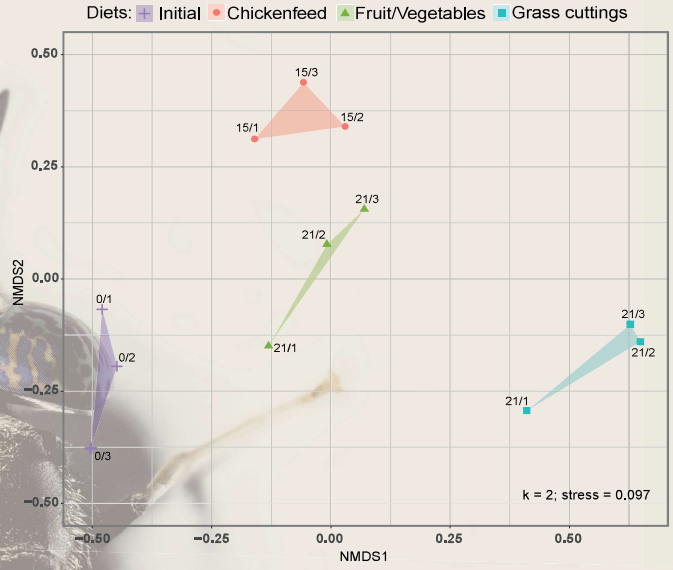

Fig 1.: NMDS from OTU data showing distances in gut communities between initial composition and compositions after exposure to different diets (values before slash indicate sampling day, after slash indicate replicate no.)

Fig 2.: Residues from feeding experiments with chickenfeed, grass cuttings and fruit/vegetable mix showing varying degradability of substrates.

Discussion

- Efficient degradation of all offered diets with similar amounts of residues.
- Conversion efficiency of organic waste to biomass was lower than that of chickenfeed.
- Development from larval to prepupal stage varied among diets depending on digestibilty of the substrate. Low-energy substrates like fruit and grass cuttings retarded larval development by 3-4 days.
- Actinobacteria, Proteobacteria and Bacteroidea accounted for high abundances in all guts, hinting towards a stable microbial core community.

Conclusion

- ▶ A **low content of organic matter** in substrates **negatively affects larval development and growth** compared with a balanced diet but results in **higher rates of substrate metabolization**.
- ▶ Similar amounts of residues coming from the degradation of various substrates distinguish the Black Soldier Fly as an **efficient all-round degrader**.
- ▶ Illumina Amplicon sequencing of bacterial gut communities in larvae raised on different diets shows **comparable development according to the type of diet** and results in **substrate-specific clusters**.

References

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